

Query Routing with Cached Responses

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Abstract

An operator serving an expensive flagship model would often be willing to answer some queries with a cheaper substitute, provided the substitute’s response is close to what the flagship would have said. We study this problem in the setting that reflects deployed practice: the available evidence is caches of raw model responses — unlabeled, unscored, and with no requirement that different models answered the same queries. We localize the product-kernel perspective space (PKPS) at each incoming query, which yields a per-query behavioral geometry over models from unpaired caches; the candidate nearest the flagship in this geometry is a good substitute, matching routing with hidden task labels while using no metadata. To offload *safely* we attach a contract: a query is served by the substitute only if the predicted deviation from the flagship is within a tolerance ε , calibrated so that $P(\text{deviation} > \varepsilon) \leq \alpha$. The required confidence signal is a kernel regression of realized same-query deviations, and we show both empirically and by a variance argument that same-query pairs are necessary for this component — but that the small paired sample every calibrated policy must already collect (candidate generations on a few hundred of the operator’s own queries; compute, not labels) is sufficient: it certifies a median 90% of traffic at ($\varepsilon=0.5, \alpha=10\%$) on a combined pool of 138 models, within five points of full-pairing and near the oracle ceiling. We measure the price of certification (a knee near $N \approx 800$ own-traffic queries, predictable in advance from query embeddings and a small pilot), the validity of the gate, and the failure mode: traffic from tasks absent from the cache silently exceeds the violation budget.

1 Introduction

Serving a frontier language model is expensive, and much of the traffic it receives does not need it. Per-token prices across a deployed fleet commonly span one to two orders of magnitude, so an operator who can identify which queries a cheaper model would answer *like the flagship* can offload those queries and keep the rest, cutting the serving bill with a bounded effect on what users see.

The evidence available for this decision is, in practice, caches of raw responses: the flagship’s own responses to logged traffic, and responses of candidate models to whatever prompts they happen to have been run on — public benchmark dumps, prior evaluations, sampled traffic. These caches are large but inconvenient. They carry no ground-truth labels or judge scores (scoring is the expensive part of evaluation; generation is not). Different models have answered different query sets, so the caches are unpaired. And they carry no reliable task taxonomy. Most prior routing work assumes at least one of these resources — per-query quality labels, task labels, or a paired evaluation matrix — that this setting does not provide.

Two problems must be solved with this evidence alone. First, *selection*: for each incoming query, which candidate would respond most like the flagship? Second, *certification*: for which queries can we bound the risk of offloading — serve the substitute only when $P(\|x_{\text{sub}} - x_{\text{flag}}\| > \varepsilon) \leq \alpha$ for an operator-chosen tolerance ε and budget α ? Selection turns out to be the easy half. Certification requires a per-query error prediction plus a calibration set, and it is here that the structure of the problem — what can and cannot be estimated from unpaired caches — becomes interesting.

We make four contributions. (1) A router built from cached responses only: a query-localized PKPS geometry supplies the substitute (no labels, no task metadata, no pairing), a kernel regression of realized same-query deviations supplies the confidence, and a conformal cutoff supplies the guarantee. (2) Evidence that the label-free pick matches routing with hidden task labels on a combined,

unlabeled pool of two evaluation suites (138 models), and an account of when selection is easy (target rank, candidate count, cache density all act through a single quantity, the neighbor-availability floor). (3) An analysis of the confidence component: its estimand decomposes so that everything except a same-query co-movement term is estimable from unpaired caches, and that term resists both plug-in and bias-corrected cross-query estimators for a variance reason — same-query differencing is the unique coupling that cancels the per-query random effect sample-wise. The practical consequence is positive: the paired sample that conformal calibration already requires, reused as regression anchors, reaches within five points of the full-pairing ceiling. (4) An operational account: the certified volume as a function of the paired-sample budget, with a knee predictable in advance from query embeddings and a small pilot; the validity of the gate as a function of calibration size; and a stated failure mode under task-level distribution shift.

2 Setting

Let $f_1, \dots, f_n$ be models with cached responses. Model f_i has answered a query set Q_i ; for $q_j \in Q_i$ we observe a response embedding $x_{ij} \in \mathbb{R}^d$ and a query embedding $u_j \in \mathbb{R}^{d_q}$. The query sets Q_i may differ arbitrarily across models, and no scores or task labels are observed. One model f_* is the flagship being served; its responses to the operator’s traffic exist by definition. For an incoming query q^* the router either serves the flagship or picks a substitute $r(q^*)$ from a candidate pool and serves it. The realized error of an offloaded query is the behavioral deviation $\|x_{r,q^*} - x_{*,q^*}\|$; we do not assume correctness scores exist, so behavioral fidelity is the objective rather than a proxy.

The operator specifies a contract (ε, α) : among offloaded queries, deviations above ε should have frequency at most α . Throughout, evaluation is leave-query-out: no model’s response to an evaluation query enters any geometry, regression, or calibration used to route that query.

3 The router

The router has three components: a pick, a confidence, and a gate.

Pick: the PKPS geometry localized at q^* . The product-kernel perspective space compares models i, i' through

$$A_{ii'} = \frac{\sum_{j \in Q_i} \sum_{l \in Q_{i'}} k_Q(u_j, u_l) k_R(x_{ij}, x_{i'l})}{\sum_{j \in Q_i} \sum_{l \in Q_{i'}} k_Q(u_j, u_l)}, \quad (1)$$

which requires no shared queries. To route we localize this comparison at the incoming query: each anchor’s contribution is weighted by $w_j = k_\sigma(u_j, u^*)$. Under a linear response kernel the localized comparison factorizes into per-model statistics: model i is represented by the weighted mean of its own cache, $\phi_i(q^*) = \sum_{j \in Q_i} w_j x_{ij} / \sum_{j \in Q_i} w_j$, and the pick is the candidate minimizing $\|\phi_i(q^*) - \phi_*(q^*)\|$. Because ϕ_i uses only model i ’s own rows, merging heterogeneous caches cannot contaminate the geometry, and fully disjoint query sets are handled without modification. The bandwidth σ is set to a fixed fraction of the median anchor-query distance; a leak-free cross-validation on anchors selects the same value (Appendix).

Confidence: deviation regression on the calibration sample. The gate needs, for each query, a prediction of the deviation the pick would realize. The estimand is a localized same-query expectation, and it decomposes as

$$\mathbb{E} \|x_{i,q} - x_{*,q}\|^2 = \|\mu_i - \mu_*\|^2 + \text{tr} \Sigma_i + \text{tr} \Sigma_* - 2 \text{tr} \text{Cov}(x_i, x_*), \quad (2)$$

with means, variances, and the covariance taken over queries near q^* . The first three terms are per-model and therefore estimable from unpaired caches — they are the PKPS quantities. The co-movement term is a joint moment: responses to the same query share a large query-driven component (models echo the prompt), and this component cancels only in same-query differences. Section 4.2 shows that estimators built without same-query pairs fail here, for variance rather than identifiability reasons. The confidence we use is the direct kernel regression of realized deviations on same-query pairs,

$$\hat{D}_i(q^*) = \frac{\sum_{j \in P_i} w_j \|x_{ij} - x_{*j}\|}{\sum_{j \in P_i} w_j}, \quad (3)$$

where P_i is the set of queries answered by both f_i and the flagship in a designated anchor sample. Crucially, this sample need not come from the caches: the flagship’s responses to the operator’s traffic exist, and running each candidate once over a few hundred logged queries is a generation cost, not a labeling cost. The same sample is required by any calibrated policy (below), so the confidence adds no assumption beyond what the guarantee itself already demands.

Gate: conformal cutoff. The operator’s own paired sample is split in half: one half provides the regression anchors P_i in (3), the other calibrates the gate. On the calibration half we observe (confidence, realized deviation) pairs; the gate offloads a query when its confidence falls below the largest cutoff whose empirical calibration violation $P(\text{dev} > \varepsilon \mid \text{offloaded})$ is at most α . Any policy that promises a deviation bound must observe realized deviations to calibrate it, and a realized deviation is by definition the flagship and a candidate answering the same query — so this paired sample is the irreducible cost of the guarantee, for every method.

4 Experiments

Data. We use two evaluation suites with Gemini-embedded queries and responses: a fully paired core of 93 models on 1,529 queries over 18 tasks, and an unpaired suite of 45 models on 6,808 queries over 16 tasks with roughly 10% incidental overlap. Their union is treated as one unlabeled pool: 138 models, 8,457 queries, 34 hidden tasks; task and suite labels are hidden from all routers and used only to define reference baselines. The flagship is the pool’s best model by anchor mean score (per suite); candidate pools are either unrestricted or restricted to models of at most 13B parameters (a serving-cost proxy).

Protocol. Unless noted, results use one protocol: 400 evaluation queries, 400 regression-anchor queries, and 400 gate-calibration queries per seed, disjoint, all excluded from every geometry; 15 seeds. Certified volumes are reported as median [IQR] across seeds because the gate’s outcome is bimodal: it either certifies its typical volume or, when the calibration draw cannot support the contract, abstains entirely — volume zero with the contract intact. Abstention rates are reported alongside. Violations are reported over certifying seeds.

4.1 The pick, and the guarantee it supports

Figure 2a evaluates selection alone on the combined pool: route each evaluation query to the candidate nearest the flagship, measure the realized deviation of the pick. The label-free PKPS pick attains mean deviation 0.378, statistically indistinguishable from routing with the hidden task labels (0.384) and well ahead of the static (non-localized) geometry (0.421) and random (0.548); the hindsight-best pick achieves 0.183. Selection difficulty is governed by a single quantity, the neighbor-availability floor — the deviation of the hindsight-best candidate: the floor and the PKPS error trace the same curve as the target’s rank varies, as the candidate pool grows, and as the cache densifies (Appendix).

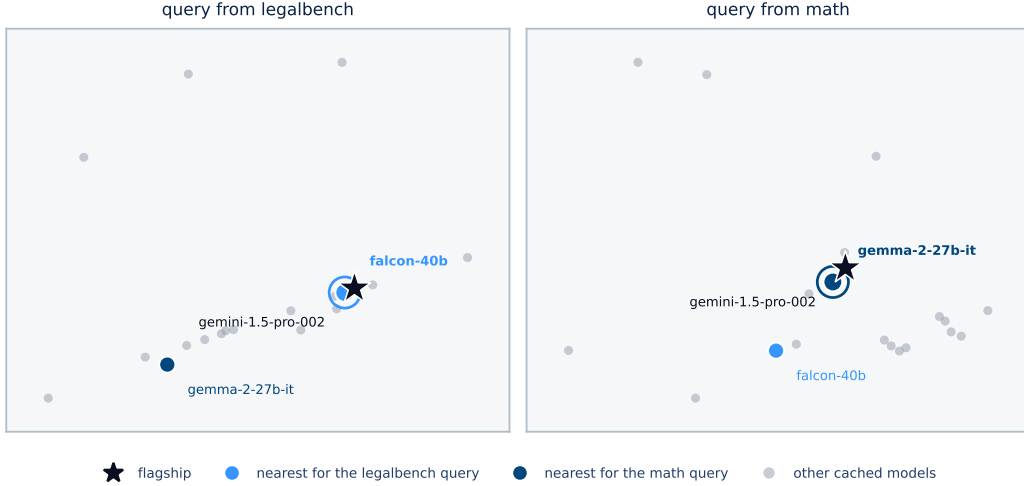


Figure 1: One response cache, two queries. The same cached models, embedded by the PKPS geometry localized at a LegalBench query (left) and a MATH query (right); the flagship is starred and both queries’ nearest candidates are highlighted in both panels. The substitute nearest the flagship changes with the query; positions are 2-D projections constrained to preserve the nearest-candidate relation.

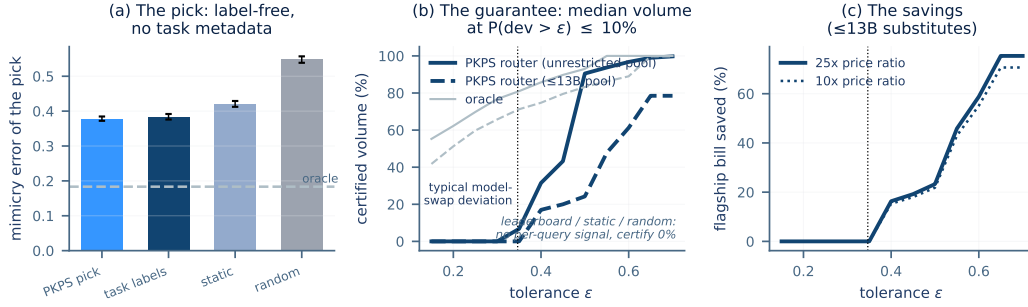


Figure 2: Does it pay? (a) Selection on the combined unlabeled pool: the label-free PKPS pick matches routing with hidden task labels; dashed line is the hindsight-best floor. (b) Certified volume at $P(\text{dev} > \epsilon) \leq 10\%$, median over 15 seeds; the oracle gates the same picks with realized deviations. Methods without a per-query error signal certify zero volume at any tolerance (Table 2). (c) Flagship bill saved on the $\leq 13\text{B}$ pool at two price ratios.

Figure 2b is the paper’s central result. At budget $\alpha = 10\%$ the router sweeps the tolerance ϵ and certifies, at each point, the median fraction of traffic it can offload with $P(\text{dev} > \epsilon) \leq \alpha$. At $\epsilon = 0.5$ it certifies a median 90% of traffic on the unrestricted pool (abstaining in 3 of 15 seeds) and 24% on the $\leq 13\text{B}$ pool; the oracle — the same gate given the realized deviations — certifies 93%. Score-free baselines do not appear on this plot because they cannot: a leaderboard pick, the static geometry, and random selection have no per-query error signal, so their violation rate at a given ϵ is a fixed population property (38–63% at $\epsilon = 0.3$; Table 2) that no volume choice can lower. The per-query contract requires a per-query error predictor. Figure 2c converts certified volume to money: at a $25\times$ flagship-to-substitute price ratio, offloading to $\leq 13\text{B}$ substitutes saves roughly a quarter of the flagship bill at $\epsilon = 0.5$ and up to three quarters at looser tolerances.

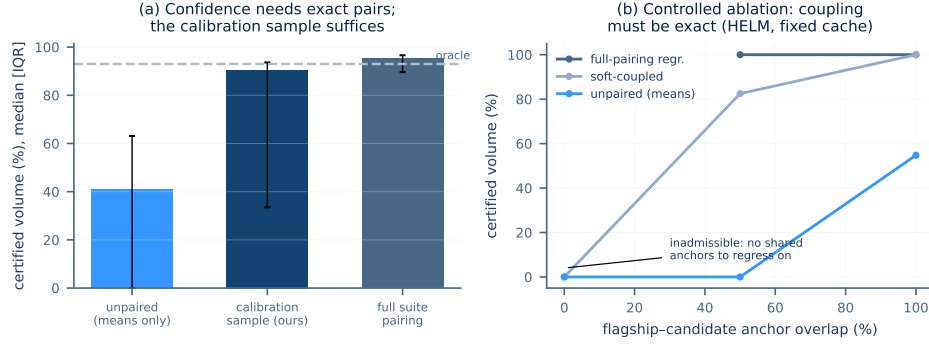


Figure 3: Why it works. (a) Certified volume by confidence signal under one protocol (median [IQR], 15 seeds): unpaired means are qualitatively insufficient; the deviation regression (3) anchored on the operator’s calibration sample reaches within five points of full-suite pairing and the oracle line. (b) Controlled pairing ablation (paired suite, fixed 50% cache per model): the regression requires shared anchors — inadmissible at zero overlap — and soft coupling over similar queries earns only its exact-match content, consistent with the variance argument in §4.2.

4.2 Why the confidence needs exact pairs, and why that is cheap

Figure 3a compares confidence signals under the single protocol above, holding the pick and gate fixed. The best fully unpaired confidence — the localized mean distance, i.e. the pick’s own geometry — certifies a median 41% with an interquartile range reaching zero (abstaining in 40% of seeds). The deviation regression anchored on the operator’s paired sample certifies 90%; anchoring it instead on all shared queries in the suites (full pairing, roughly 1,400 shared anchors per pair against the flagship) certifies 95%. The oracle line is 93%. The gap that matters is the first one: means-only confidence is qualitatively insufficient, and the decomposition (2) says why — it omits both variance terms and the co-movement term, and the latter is the dominant contributor because responses echo their prompts.

Two natural attempts to recover the co-movement term without pairs fail, and the failure is informative. The plug-in bound that sets $\text{Cov} = 0$ certifies approximately zero: the omitted term is most of the variance, not a correction. A soft-coupled estimator that differences responses to *similar* queries, with the query-difference penalty explicitly removed using within-model distance curves (an unbiased correction under an additive query effect), drowns: it subtracts two nuisance-dominated quantities to extract a small signal, and its ordering degrades below chance. The general statement is a variance argument. The query effect $g(q)$ is a per-query random effect that dominates response variation; same-query differencing is the unique coupling that cancels it sample-wise rather than in expectation, so every cross-query estimator carries variance at least the g -variability at the closest available query match. Benchmark caches contain no near-duplicate queries, so that variance exceeds the model-difference signal. This is the matched-pairs design argument: matching removes the subject effect, and these data contain no twins. Figure 3b shows the controlled version: with per-model cache size fixed, the paired regression’s certified volume falls as flagship–candidate anchor overlap drops from 100% to 50% and is undefined at 0%, while the soft-coupled estimator earns exactly its exact-match content and nothing more.

The positive consequence: because certification already forces the operator to collect realized deviations — candidate generations on a few hundred of their own queries — the necessary pairing is already in hand. Equation (3) anchored on that sample is within five points of the full-pairing ceiling, and on-traffic anchors are worth several times their count in suite-wide anchors (Appendix).

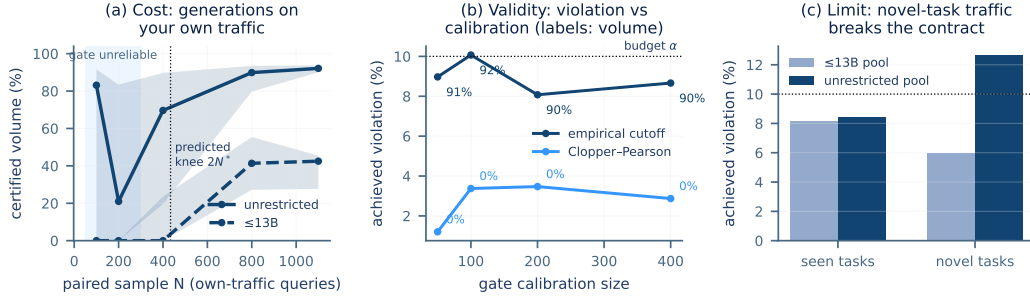


Figure 4: Cost and limits. (a) Certified volume vs. the paired-sample budget N (median [IQR], nested subsets, 10 seeds); the region below $N \approx 300$ is shaded because the gate itself is unreliable there; dotted line marks the predicted knee $2N^*$. (b) Achieved violation vs. gate-calibration size for the empirical and Clopper-Pearson cutoff rules (labels: median certified volume; the conservative rule mostly abstains). (c) Traffic from tasks absent from the cache exceeds the violation budget on the unrestricted pool; the guarantee is conditional on exchangeability.

Table 1: The confidence ladder under one protocol (combined pool, unrestricted candidates, $(\epsilon=0.5, \alpha=10\%)$, 15 seeds). Volumes are medians; violations are over certifying seeds.

confidence	coupling	pairing assumed	volume	abstain	viol.
localized means	none (unpaired)	none	41%	40%	.11
deviation regr. (ours)	exact	calibration sample only	90%	20%	.11
deviation regr. (full)	exact	all shared suite queries	95%	0%	.09
oracle	—	(peeks at eval)	93%	0%	.11

4.3 Cost, validity, and limits

Figure 4a prices the paired sample. Sweeping the total budget N (half regression anchors, half gate calibration, nested within seed so the comparison is paired), certified volume on the unrestricted pool reaches its plateau near $N \approx 800$ own-traffic queries; below $N \approx 300$ the gate itself is unreliable and outcomes are bimodal. The knee is predictable before collecting anything: the confidence’s signal-to-noise crosses one at $N^* = s^2/(\kappa\tau^2)$, where the kernel coverage κ is computable from query embeddings alone and the noise and signal scales s^2, τ^2 from a pilot of roughly one hundred pairs; on these data $N^* \approx 217$ anchors, matching the observed knee within a factor of two, with the calibration side adding a few hundred more.

Figure 4b examines the gate rule. The empirical cutoff tracks the budget from above as calibration grows (violations 8–10% at volumes near 90%); a Clopper-Pearson cutoff is safe at every size but abstains in most seeds at these calibration sizes — a provable gate needs either more calibration data or a weaker correction. Figure 4c states the failure mode. When evaluation traffic comes from tasks entirely absent from the cache, the confidence distribution barely shifts (AUC 0.62 for detecting novelty — too weak to gate on), the gate offloads novel-task queries at a normal rate, and the achieved violation exceeds the budget (12.6% against 10% on the unrestricted pool). The guarantee is therefore conditional on traffic exchangeability with the calibration sample; detecting out-of-cache queries is an open problem, and the natural coverage statistics (effective local sample size) are anti-correlated with novelty on these data.

5 Discussion

The two halves of the router obey different laws. The pick — a first moment of the localized product kernel — factorizes over models, needs no pairing of any kind, and is robust to merging arbitrary caches; this is the regime where the PKPS construction, comparing responses to similar

Table 2: Methods without a per-query error signal cannot hold a contract: their violation rate at tolerance ε is a fixed population property, independent of how much traffic is offloaded (combined pool, eval split).

ε	$\leq 13\text{B pool}$			unrestricted		
	0.1	0.3	0.5	0.1	0.3	0.5
leaderboard pick	70%	46%	28%	63%	38%	19%
random pick	77%	56%	39%	71%	48%	32%

rather than identical queries, does exactly what it was designed to do. The guarantee — a second moment — does not factorize: predicting a same-query difference requires observing same-query differences, for variance rather than identifiability reasons. What makes the method practical is that the requirement collapses onto a sample the guarantee already charges for. We read this as a general design principle for routing systems built on cached responses: spend the generation budget on your own traffic, not on more benchmark coverage — on-traffic anchors were worth several times their count in suite anchors throughout our experiments.

Three limitations are explicit in the results. The contract is conditional on traffic exchangeability, and novel-task traffic breaks it silently; until an out-of-cache detector exists, the guarantee should be stated for in-distribution traffic. The empirical gate rides the budget from above and occasionally abstains; a provably valid gate at these calibration sizes remains open. And the behavioral objective bounds deviation in an embedding space; where correctness labels exist, deviation is a proxy whose adequacy should be checked (on our suites, behavioral policies held 95–99% of flagship score at conservative budgets; Appendix).